Smart MCQ Solver Challenge

Milestone 2 Report

Pre-trained transformers — tokenization, embeddings, attention, semantic similarity & zero-shot classification

DATASET	smart-mcq-solver-challenge — train.csv (2000 rows, 8 cols)
TASK	5-way multiple-choice question answering (options A–E)
MODELS	bert-base-uncased, all-MiniLM-L6-v2, flan-t5-small
SCOPE	13 checkpoint questions

1. Environment & data preparation

The training set is loaded into a Hugging Face `Dataset` and a `combined_text` field is created by concatenating each prompt with its option A.

Q1 Length of combined prompt+option text at row 51

What it does. Measures the character length of the merged `prompt + ' ' + A` string for sample 51.

How. Built with a `Dataset.map` over the training set, then read `ds[51]['combined_text']`.

RESULT **614 characters**

2. BERT tokenizer internals

Q2 BERT vocabulary size

What it does. Reports the size of the WordPiece vocabulary used by `bert-base-uncased`.

How. Loaded the tokenizer and read `tokenizer.vocab_size`.

RESULT **30,522 tokens**

Q3 Token ID of the [SEP] separator

What it does. Finds the integer ID BERT assigns to the special separator token.

How. `tokenizer.convert_tokens_to_ids('[SEP]')`.

RESULT **102**

Q4 Shape of a batched, padded input tensor

What it does. Tokenizes all 2000 prompts to a fixed length and inspects the tensor shape.

How. Padded to `max_length=128` with truncation; read `input_ids.shape`.

RESULT **(2000, 128) — 2000 prompts × 128 tokens**

3. Model dimensions & forward pass

Q5 Per-head attention dimension

What it does. Computes the size of each attention head in BERT-base.

How. Hidden size 768 split across 12 heads: `768 // 12`.

RESULT **64**

Q6 Shape of the last hidden state

What it does. Runs one prompt through the model and inspects the contextual embedding tensor.

How. Forward pass under `torch.no_grad()`; the 31-token prompt yields `last_hidden_state`.

RESULT **(1, 31, 768) — batch 1, 31 tokens, 768-dim**

Q7 Sum of the first five [CLS] embedding values

What it does. Takes the [CLS] sentence vector (token 0) and sums its first five components.

How. Sliced `last_hidden_state[0, 0, :5]`, summed, rounded to 4 dp.

RESULT **-1.2001**

4. Attention analysis

Q8 Attention weight from [CLS] to 'fusion'

What it does. Extracts how much the [CLS] token attends to ‘fusion’ in a sample sentence.

How. Reloaded BERT with `output_attentions=True`; last layer, first head, [CLS] → ‘fusion’.

RESULT **0.1025**

5. Semantic similarity (Sentence-Transformers)

Q9 Cosine similarity between prompt and option B

What it does. Scores how semantically close the row-0 prompt is to its option B.

How. Encoded both with `all-MiniLM-L6-v2`; applied `util.cos_sim`.

RESULT **0.7658**

Q10 MiniLM retrieval baseline (MAP@3)

What it does. Ranks the five options per question by embedding similarity to the prompt and scores the ranking.

How. Two pipelines — TF-IDF and MiniLM — each producing a top-3, evaluated with MAP@3. Count = questions MiniLM ranked differently from TF-IDF.

RESULT **MiniLM MAP@3 = 0.4231 | Count = 462**

6. Zero-shot classification & generation

Q11 Top zero-shot probability (softmax)

What it does. Treats the three options as candidate labels; asks a zero-shot classifier which best matches the prompt.

How. `pipeline('zero-shot-classification')` in softmax mode; top score. The three probs sum to 1.0.

RESULT **0.4575**

Q12 Softmax vs. sigmoid probability-mass gap

What it does. Contrasts single-label and multi-label scoring for the same question.

How. Re-ran with `multi_label=True` (sigmoid); absolute difference of summed probs (1.0 vs ~0.0005).

RESULT **0.9995**

Q13 Generative MCQ baseline (Flan-T5)

What it does. Checks whether a small instruction-tuned seq2seq model picks the right option directly.

How. Formatted row 0 as an A/B question; generated with `flan-t5-small` (`max_new_tokens=5`, greedy).

RESULT **`Prediction: "B"`**

Answer summary

Q	Checkpoint	Answer
1	combined_text length at index 51	**614**
2	BERT vocabulary size	**30,522**
3	[SEP] token ID	**102**
4	Padded input_ids shape	**(2000, 128)**
5	Attention head dimension	**64**
6	last_hidden_state shape	**(1, 31, 768)**
7	Sum of first 5 [CLS] values	**-1.2001**
8	Attention [CLS] -> fusion	**0.1025**
9	Cosine similarity (prompt, B)	**0.7658**
10	MiniLM MAP@3 / Count	**0.4231 / 462**
11	Top zero-shot prob (softmax)	**0.4575**
12	|sum softmax - sum sigmoid|	**0.9995**
13	Flan-T5 prediction (row 0)	**B**

Takeaway. Milestone 2 exercises the moving parts of a pre-trained transformer — tokenization, embeddings, attention and similarity — and benchmarks three zero-shot / retrieval baselines for the MCQ task before any fine-tuning.